

The Carrier Geometry of the Critical Line: A Worked Example with Two Conditional GRH Routes

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Abstract

The classical L -function is a one-dimensional readout of a three-dimensional object. This paper runs that frame on the simplest configuration it admits — the $\pi/3$ carrier with an ordinary Dirichlet L -function riding it — as an exact audit of the geometric midpoint, the native focal events, and the interface needed to identify carrier events with the analytic zero set. The unconditional content: the helix midpoint is a geometric *output* (the unique area-law scale balance and the no-drift law), a completed native 3D zero is an exact focal cancellation at physical height $Z > 0$ whose chart ordinate is $\log Z$, the finite-bank detector is exactly equivalent to harmonic-pencil rank drop, and both Gram operators admit spatial lifts $G \otimes I_3$ that preserve and reflect rank drop. Two conditional GRH routes are retained under the identity and the correlation readings, each with its exact premises displayed and compiler-audited. The genuine operator reduction is compiled: GRH follows for any Hermitian tower whose canonical eigenvalue resolvent traces converge locally uniformly off the real axis to the completed channel (`grh_of_hermitian_traceLimit`); the earlier uncompleted typing is proved uninstantiable by any operator (`no_selfAdjoint_uncompleted_capture`); and the one open box — windowed exhaustion of the compiled ledger — is proved *equivalent* to Mathlib’s RiemannHypothesis (`rh_iff_symmetrizedLineTower_traceLimit`): the wall neither overshoots nor undershoots RH. This paper does not prove RH or GRH; it isolates them to a single named limit, machine-checked from both sides.

Three registers are used and never blended. *Proven*: machine-checked in Lean 4, every named theorem at axiom footprint `{propext, Classical.choice, Quot.sound}` with no sorry, built in-tree; the Lean sources live with the main paper [1] and are named at each use. *Measured*: precision-tracking numerics — calibration and independent reproduction of proven statements, never their proof. *Conditional*: the two GRH routes are displayed with their exact premises, and the compiler audit identifies which premise supplies the critical-line conclusion. No claim in this paper exceeds its register. The falsifier posture is inherited from the main paper [1]: the working dichotomy is stated with its disconfirmation criteria in print, and searches against the favored branch are part of the program, not a threat to it.

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54 1 Overview

55 This paper runs the machinery of the main development [1] on the simplest configuration it admits
56 — the $\pi/3$ carrier with an ordinary Dirichlet L -function riding it. Its purpose is an exact audit of the
57 geometric midpoint, the native focal events, and the additional interface needed to identify those
58 events with the analytic zero set.

59 Throughout, the hypothesis is stated in Riemann's own terms, and the nuance matters. Riemann
60 did not conjecture about a decimal. Substituting $s = \frac{1}{2} + ti$ into the completed zeta function and
61 regarding the result $\xi(t)$ as an entire function of t , he counted its real roots up to a given bound and
62 wrote [2]:

63 “... es ist sehr wahrscheinlich, dass alle Wurzeln reell sind. Hiervon wäre allerdings ein
64 strenger Beweis zu wünschen; ich habe indess die Aufsuchung desselben nach einigen
65 flüchtigen vergeblichen Versuchen vorläufig bei Seite gelassen, da er für den nächsten
66 Zweck meiner Untersuchung entbehrlich schien.”

67 (“... it is very probable that all the roots are real. One would of course like a rigorous
68 proof of this; I have for the time being, after some fleeting futile attempts, provisionally
69 set the search aside, as it appears dispensable for the immediate objective of my
70 investigation.”)

The hypothesis, as its author stated it, is that the roots are *real*: that they lie on the *real axis* of the ξ chart. The familiar “ $\text{Re } s = \frac{1}{2}$ ” is the same statement after the change of variable, and the half in it is not a privileged decimal value. It is `UNIT/2`: the midpoint of the unit-width frame that the functional equation $s \leftrightarrow 1 - s$ reflects — in projection geometry, the midline of the frame; on the carrier, the weld axis fixed by the involution J . On the raw carrier this midpoint is not stipulated: the arclength area law first proves that the unique scale-balanced fiber exponent is $\frac{1}{2}$, and no-drift conservation then fixes the native source modes there (§2.5). Riemann’s real axis, the classical midline, and the carrier’s weld axis are one locus in three charts, and we will use whichever name keeps a given argument honest. Throughout this paper, GRH means: for every modulus q and every Dirichlet character $\chi \bmod q$, all roots of the completed $\xi(t, \chi)$ are real — equivalently, every nontrivial zero of $L(s, \chi)$ lies on the midline. The principal case $q = 1$ is the Riemann zeta function, and each proof specializes there to the Riemann Hypothesis.

The fibre is the three-dimensional representation of its configured L -function, not a surrogate for it — and this is now a packaged *relational-equality theorem*, not a pair of convergence statements. The capstone family (`DirichletFiberEquivalence`, `UniversalFiberEquivalence`, `PrincipalFiberEquivalence`) proves, unconditionally: for every nonprincipal character the fibre readout converges throughout $\text{Re } s > 0$ to the analytically *continued* $L(s, \chi)$, and the classical L -function is *the unique entire extension* of that readout — any entire function reading the fibre on the strip equals $L(\cdot, \chi)$ on all of $\mathbb{C}$ (`dirichletFiberEquivalence`, `packaging spectralFiber_readout_tendsto_LFunction` with the identity-theorem `core entire_eq_of_eqOn_isOpen`). The uniqueness core is class-free: *any* coefficient bank admits at most one entire reading (`entire_readings_unique` — no character, no growth condition, no functional equation consumed). The principal character rides the eta conduction: ζ is recovered from the rescaled fibre readout wherever $1 - 2^{1-s} \neq 0$, with the pole booked as the conduction factor’s zero at $s = 1$ and the alternating window cancelling exactly per adjacent pair (`principalFiberEquivalence`, `packaging etaSpectralFiber_readout_recovers_zeta`). Alongside: conduction is lossless, the ledgered projection exact, and vanishing transports exactly — the completed focal residual vanishes at height Z iff the represented L -value does. The state is three-dimensional, while its analytic readout is complex-valued; the equivalence eliminates *representation* as an interface question — what remains between the fibre and the analytic zero set is coverage, never identity of the represented function.

The Lean development then separates three coordinates and statements that must not be conflated. First, the unit-gauge Archimedean area law has a unique scale-balanced abscissa, denoted σ_* , and proves $\sigma_* = \frac{1}{2}$. Second, a completed native 3D zero is an exact focal cancellation at a positive physical helix height Z . Its chart ordinate is not Z , but

$$y = \log Z, \quad \text{carrierPointAtHeight}(Z) = \sigma_* + i \log Z.$$

Third, identifying every analytic zero with one of these carrier events is a separate zero-set coverage statement.

The two proof bodies below are retained as two conditional routes under two readings: the identity reading and the correlation reading. The correction is an exact account of what each displayed premise supplies. The identity route assumes analytic-zero identity with a native event; the correlation route assumes a pointwise chart-zero-to-kernel correspondence. Neither premise is now described as a consequence of the internal evidence dossier alone.

This separation is now enforced in the source. The internal ordinate chart `carrierPoint y` uses `carrierAbscissa`, not a literal half; the physical-height interface is `carrierPointAtHeight Z`; `carrierAbscissa_eq_half` derives its normal form from `carrierScaleBalanced_iff`. The midpoint-free proposition `NativeCarrierCoverage( $\chi$ )` says that the zero of ordinate $\Im \rho$ produces

116 a completed focal event at physical height $Z = e^{\Im \rho}$. Finally, the compiled decomposition theorem

$$\text{threeDCrossings_iff_grh_and_nativeCarrierCoverage}$$

117 proves the exact content of the older identity premise:

$$\text{ThreeD_crossings_are_real_zeros}(\chi) \iff \text{GRH}(\chi) \wedge \text{NativeCarrierCoverage}(\chi).$$

118 Accordingly, that older premise is retained as a legacy conditional interface, not presented as an
119 independently established bridge from analytic zeros to carrier events.

120 **Operator audit.** At physical height $Z > 0$, put $\gamma = y = \log Z$. The local fibre operator used by the
121 elementary 3D calculation is

$$\text{vonNeumannOp}(\gamma) = \gamma \text{ id}_{\mathbb{C}}.$$

122 Its symmetry is a valid theorem, but it is a theorem about a one-dimensional scalar fibre. Likewise,
123 $\text{specBchan}(\gamma, s) = \gamma + i(s - \sigma_*)$ vanishes precisely when $s = \sigma_* + i\gamma$. These facts prove that a point
124 already coupled to that fibre has the carrier's derived abscissa. They do not prove that the nontrivial
125 zeros of $L(s, \chi)$ exhaust, or are exhausted by, those fibre events. This paper therefore calls the scalar
126 operator a *local carrier coordinate detector*, not a global Hilbert–Pólya operator.

127 There is now also one fixed operator on the full geometric 3D carrier, defined before a character
128 or event predicate is introduced. For fixed nondegenerate carrier parameters (p, r) with $p \neq 0$, Lean
129 defines

$$C_{p,r} = \{(X, y) : X = \gamma_{p,r}(y)\}, \quad \mathcal{H}_{p,r}^{(3)} = C_{p,r} \rightarrow_0 \mathbb{C},$$

130 and

$$A_{p,r}^{(3)} = \text{diag}_{s \in C_{p,r}}(y_s) \quad (\text{carrierThreeDOperator}).$$

131 The state type `CarrierState3D` stores the spatial point X , its logarithmic ordinate y , and only the
132 geometric equation $X = \text{gammaY}(p, r, y)$. It contains no character, L -function, zero predicate, focal
133 cancellation, or completed-event certificate. Lean proves that the third coordinate is $e^y > 0$, that
134 $A_{p,r}^{(3)}$ is symmetric (`carrierThreeDOperator_isSymmetric`), and, for every geometric carrier state s ,
135 that $\delta_s \neq 0$ satisfies

$$A_{p,r}^{(3)} \delta_s = y_s \delta_s \quad (\text{carrierThreeDOperator_eigenvector}).$$

136 A completed event e is afterward mapped by `CompletedThreeDEigenEvent.toCarrierState` to
137 the ambient state with $y_s = \log Z_e$. Lean proves that this state's third spatial coordinate is exactly
138 the original Z_e and that its basis vector is an eigenvector of the operator already defined on all carrier
139 states (`carrierThreeDOperator_completedEvent_eigenvector`). Thus the event certificate selects
140 a state; it is not used to construct the operator or its domain. The older `completedThreeDOperator`
141 is retained as the restriction indexed only by completed events for API compatibility.

142 The Gram pencils also carry an explicit three-coordinate compatibility layer. For either two-
143 channel Gram matrix G , Lean defines

$$\text{spatialLiftGram}(G) = G \otimes I_3 \quad \text{on} \quad \mathbb{C}^2 \otimes \mathbb{C}^3$$

144 and proves

$$\det(\text{spatialLiftGram}(G)) = (\det G)^3, \quad \det(\text{spatialLiftGram}(G)) = 0 \iff \det G = 0.$$

The physical-height definitions `completedHarmonicGram3DAtHeight` and `vonNeumannGram3DAtHeight` instantiate this lift. The theorem

`completedThreeDZeroAtHeight_twoGram3D_rankDrop`

sends every completed 3D focal event to both lifted rank drops. Thus adjoining the three spatial carrier coordinates cannot create a new vanishing or erase an existing channel cancellation. This establishes 3D compatibility of the local pencils. The ambient carrier operator above then reads their marked crossings inside a state space that already existed; the spatial lifts and the event embedding perform distinct jobs.

Zero-set interface. The completed native event is `CompletedThreeDZeroAtHeight` χ Z , definitionally positivity of the physical height together with exact vanishing of the completed focal residual. Lean proves

$$\begin{aligned} & \text{CompletedThreeDZeroAtHeight } \chi \ Z \\ & \iff L(\text{carrierPointAtHeight}(Z), \chi) = 0, \quad Z > 0, \end{aligned}$$

by `completedThreeDZeroAtHeight_iff_L_zero`. Thus a completed 3D crossing is exactly a genuine zero of the represented L -function at its own readout point. The separate finite object `ThreeDZero` χ bank y is the truncated ordinate-level focal detector and must not be substituted for the completed event without its limiting bridge. An arbitrary analytic event is membership $\rho \in \text{NontrivialZeros}(\chi)$, which includes $L(\rho, \chi) = 0$ in the critical strip. The coverage step asks when that analytic fiber projects to the readout of a physical-height event.

For the principal channel this coverage is now decomposed by a compiled zero-to-source transfer rather than left as a bare identification. The constructor `principalZeroAnalyticFiber3D` sends every genuine $\rho \in \text{ZD.NontrivialZeros}$ to the literal eta-configured three-dimensional `Vec3` fiber at ρ , proves that its rescaled plane partial sums converge to $\zeta(\rho) = 0$, and supplies the event-independent ambient helix eigenstate at ordinate $\Im \rho$ and physical height $e^{\Im \rho}$. This first transfer does not mention a midpoint. Before any equality between the analytic parameter and a carrier readout is requested, the constructor `principalZero_focalCancellation_on_carrier` registers the analytic closure as a `PrincipalZeroNativeCarrierEvent3D`. The event contains a typed `CarrierState3D` on the Archimedean helix at that same physical height. Its theorems `.on_archimedeanHelix`, `.physicalHeight_eq_focalHeight`, and `.ambient_eigenvector` prove literal helix membership, equality of marked heights, and the ambient 3D eigenvector equation. Its `.noRadialDrift` theorem follows directly from helix membership through `carrierPointAtHeight_noRadialDrift`; neither a Euclidean fiber radius nor a radial-limit premise is used. The area law is applied only afterwards by `.carrierReadout_scaleBalanced`. The constructor `principalZeroNative3DSourceTransfer_of_noRadialDrift` then performs the geometric projection into a conservative source and the complete native focal/operator certificate from the area-law equation $n^{\text{Re } \rho - \sigma_*} = 1$. Its converse is proved from source conservation, giving the exact theorem `nonempty_principalZeroNative3DSourceTransfer_iff_noRadialDrift`. Thus the native landing is not encoded by writing $\rho = \frac{1}{2} + iz$; the half-unit is the evaluated value of the area-law-selected σ_* after no drift has been established.

Here “fiber rides carrier, carrier rides helix” is a typed composition. The first map is `PrincipalZeroAnalyticFiber3D.RidesCarrier`, equality of the literal $\mathbb{N} \rightarrow \text{Vec3}$ analytic fiber with the spectral fiber evaluated at its carrier-height readout. The second is the independent geometric field `CarrierState3D.point_eq_gammaY`. Lean proves that the first map is equivalent to the parameter-preserving weld by `ridesCarrier_iff_carrierWeld`, and every completed native

transfer proves both maps simultaneously in `fiberRidesCarrier_carrierRidesHelix`. Merely sharing an ordinate is not used as a substitute for the fiber attachment.

The exact completed focal event has carrier support by construction and theorem, not by a later chart assumption. Namely, `CompletedFocalCancellationRepresents( $\chi, \rho$ )` requires a positive physical height Z , exact vanishing of $D_{\text{cell}} \chi Z$, and the same-parameter readout `carrierPointAtHeight  $Z = \rho$` . Lean proves `completedFocalCancellationRepresents_iff`:

$$\text{CompletedFocalCancellationRepresents}(\chi, \rho) \iff L(\rho, \chi) = 0 \wedge \text{Re } \rho = \sigma_*.$$

Thus `no_offCarrier_completedFocalCancellation` rules out an off-carrier focal cancellation pointwise, before the $S(t)$ ledger or a resolvent trace is formed.

The function $S(t)$ is the global coordinate map for this readout, not a local numerical coincidence. In the compiled registration law,

$$N_{\pi/3}(e^t) - N_1(e^t) = S(t),$$

and `count_hasJump_iff_S_hasJump` identifies its jumps with the global analytic event ledger, including multiplicity through `residue_eq_Smult_jump`. Thus $S(t)$ globally transports the carrier-scale census to the unit-chart census. This establishes the global coordinate bookkeeping used in the correlation route. Pointwise, the compiled transfer theorem above identifies native landing with no drift; globally, `allNative3DSourceTransfer_of_selfAdjointXiReceiver` selects that landing for every nontrivial zero in both half-planes. Moreover the contour defect cannot hide an off-carrier contribution: `classicalSContour_sub_Smult_pos_of_offCarrier` proves that any enclosed off-carrier zero makes `classicalSContour - Smult` > 0 , with lower bound its positive analytic order. At the operator level, `zeroPoleResolvent_eq_carrierBasisReadout_iff_scaleBalanced` proves basiswise that the analytic pole receiver equals the independently defined event-free carrier receiver exactly at area-law balance; inversion is injective, so neither a trace sum nor a reflected partner can create this equality. The corresponding global equivalence is `globalCoordinateIdentification_iff_eachZeroPoleResolvent_eq_carrierBasisReadout`. The summed bridge is now derived only after this pointwise comparison. The definitions `xiZeroResolventKernel` and `xiCarrierHeightResolventKernel` retain the zero basis index, analytic multiplicity, and Hadamard counterterm, while `xiZeroResolventKernel_eq_carrierHeightKernel_iff_scaleBalanced` proves their equality exactly at area-law balance. Lean then compiles `globalCoordinateIdentification_iff_basisResolvedCarrierResolventBridge`; after its basiswise equality, `carrierHeightResolventTrace_of_globalCoordinateIdentification` rewrites the completed xi-channel as the regularized trace of the real carrier-height receiver. No equality of summed traces is used to manufacture the weld of an individual analytic mode. At the state level, `nonempty_principalZeroNative3DSourceTransfer_iff_fiberRidesCarrier` proves for each zero that a completed native source exists exactly when the literal analytic $\mathbb{N} \rightarrow \text{Vec3}$ fiber equals the carrier-readout fiber at its own physical focal height. Its quantified form `allNative3DSourceTransfer_iff_allCanonicalZeroFibersRideCarrier` and the operator form `allNative3DSourceTransfer_iff_zeroLedger_eq_carrierHeight` make the source/state/operator weld explicit, rather than treating the common ordinate as sufficient. The operator propagation is then literal on the analytic-zero basis: `xiZeroLedgerOperator_eq_carrierHeight_of_allNative3DSourceTransfer` uses each completed transfer to prove that its zero-ledger coefficient equals its real carrier-height coefficient. Consequently `globalCoordinateIdentification_of_allNative3DSourceTransfer` returns `classicalSContour = Smult` at every good height, without a summed-trace cancellation step. The receiver capstone `globalCoordinateIdentification_and_zeroLedger_eq_carrierHeight_of_selfAdjointXiReceiver` returns the global coordinate identity and the operator equality together. Under the stated (Z -3D) reading that a native

229 3D zero is primary and its 1D analytic zero is its shadow, this assembly is now exposed
 230 by one theorem: `globalCoordinateIdentification_basisResolvedBridge_and_carrierTrace_`
 231 `of_allNative3DSourceTransfer`. Its sole conditional input is one completed native source transfer
 232 for every analytic shadow. It returns simultaneously the global coordinate identity, the equality of
 233 the two resolvent kernels on every zero basis mode, and the regularized carrier-height trace. Thus
 234 the shadow conditional is used once at the source/state weld; no equation $\text{Re } \rho = 1/2$ is supplied as
 235 an additional premise to the capstone.

236 Formal dependency table.

	Statement	Proven here	Logical role
	<code>carrierAbscissa = $\frac{1}{2}$</code> <code>PrincipalZero</code> <code>AnalyticFiber3D(ρ)</code>	yes, from the area law yes, for every genuine zeta zero	fixes the native carrier image literal 3D analytic state, zero readout, physical height and ambient eigenvector
	<code>PrincipalZero</code> <code>NativeCarrierEvent3D(ρ)</code>	yes, for every genuine zeta zero	registers focal closure on a typed helix state at the same height; no drift follows from carrier membership, with no radius comparison
237	<code>PrincipalZero</code> <code>Native3DSourceTransfer(ρ)</code>	yes, from no drift; converse proved	connects the analytic fiber to its conservative source and complete focal certificate without a midpoint input
	<code>ThreeD_crossings_are</code> <code>real_zeros(χ)</code>	legacy premise	equals $\text{GRH}(\chi) \wedge$ <code>NativeCarrierCoverage(χ)</code>
	<code>OneD_zeta_zero</code> <code>correlated(χ)</code>	yes, by its right branch	routing only; it supplies no kernel identification
	<code>OneDCorrelation</code> <code>Evidence(χ)</code>	a separate premise	places a scalar-fibre kernel at every analytic chart zero and by itself implies the critical-line conclusion

238 **What the supporting facts establish.** The finite phasor, Gram, projection, counting, and convergence
 239 theorems certify the internal carrier mechanics and several chart-side audits. A diagonal operator
 240 indexed by already located analytic zeros checks counts and multiplicities; it does not locate those
 241 zeros. Similarly, local identities between a focal residual and an L -value must be read with their
 242 stated domains and limiting hypotheses. None of these facts is used below as a substitute for the
 243 analytic-zero-to-native-event bridge.

244 **Scope.** The midpoint theorem is independent of $L(s, \chi)$: it is a geometric statement about which
 245 fibre scaling remains finite and drift-free on the Archimedean helix. The GRH conditionals are
 246 different statements about whether analytic zeros are represented on that geometric image. The
 247 revised Lean declarations and theorems expose this boundary rather than asking the reader to infer
 248 it from prose.

249 2 The carrier construction and its interfaces

250 Everything in this section is main-development machinery [1] specialized to the basic configuration;
 251 every object carries a pointer to its general construction. The objects come first — the state space,

the carrier that lives in it, the fiber that rides the carrier — then the maps between scales, then the events, then the operators that read them.

2.1 The 3D state space

The ambient space is three-dimensional, with named coordinates: *radius* r , *phase* θ , and physical *height* Z . A complete carrier coordinate state is a point

$$X = (r, \theta, Z) \in \mathbb{R}_{\geq 0} \times \mathbb{R} \times \mathbb{R}_{\geq 0},$$

and this is the space in which everything below lives: the carrier is a curve in it, a phasor is a vector in the transverse (r, θ) -plane at its height, and the *bank state* at height Z is the accumulated sum of every phasor that has entered below Z — one such state per lane. The 3D representation of $L(s, \chi)$ is exactly this pair of lane states, evolving continuously as Z grows; the operators of §2.8 read it, and the events of §2.7 are its alignments.

Height is the distinguished coordinate. It is the one coordinate the 1D chart retains (§2.4), every event in this paper is indexed by the height at which it occurs, and there is no convergence gate anywhere in the space: states exist and evolve at every height, and the half-plane $\operatorname{Re} s > 1$ of the classical series is a property of the compressed chart, not of the state space.

2.2 The carrier: a double-ended helix on the rescaled number line

The carrier is the source-independent geometric object in the state space — fixed, by the source-independent double-carrier theorem of [1], before any character is attached. It is double-ended: two lanes $C = C_+ \sqcup C_-$ over a common raw parameter, realized as

$$\Gamma_{\pm}(t) = (R(t) \cos(\pm\Theta(t)), R(t) \sin(\pm\Theta(t)), Z(t)),$$

with the global involution $J : C_+ \leftrightarrow C_-$, $J^2 = \operatorname{id}$, exchanging the lanes while preserving radius and height and reversing the transverse phase. The plus lane is the helix; the minus lane is its conjugate anti-helix at the same height. In the basic configuration the helix is Archimedean with constant pitch of one unit — one unit of height per turn, the transverse shadow an Archimedean spiral — so the winding, radius, and height laws grow together with no free parameter.

The carrier admits an elementary description: it is the positive number line, rescaled. The height coordinate *is* the number line — the integer n sits at height $Z = n$ — and the rescaling is harmonic: a unit step of the integers advances the winding by $\pi/3$. Six steps close one full turn, so the finite harmonic cells are the μ_6 buckets, and the natural coefficient ring of the closure arithmetic is the Eisenstein ring $\mathbb{Z}[\zeta_6]$, in which the cancellation at a closed cell is exact — integer arithmetic, no limit taken (the primary-construct section of [1]). Nothing about any source is encoded in this choice; it is the harmonic scale at which the carrier's cells close exactly, and every integer occupies the same carrier whether or not any character ever reads it.

The two lanes are not decorative. The functional equation of the readout is the involution J read phasor by phasor: on the weld axis, height inversion coincides with conjugation, and J supplies exactly that exchange (the carrier reflection theorem and bank reflection lemma of [1]). What the classical theory obtains from Poisson summation, the carrier carries as a geometric symmetry of the state space, fixed before any source is attached. The weld axis — the fixed locus of J — is the carrier's copy of Riemann's real axis: the midline at $\operatorname{UNIT}/2$, derived from the arclength area law and no-drift conservation in §2.5, then expressed in the unit-width chart of §1. Figure 1 shows where the origin and the functional equation enter: the double-ended object welded at $z = 1$, the height inversion pivoting there, and J supplying it as conjugation on the axis.

the double-ended object: origin and functional equation

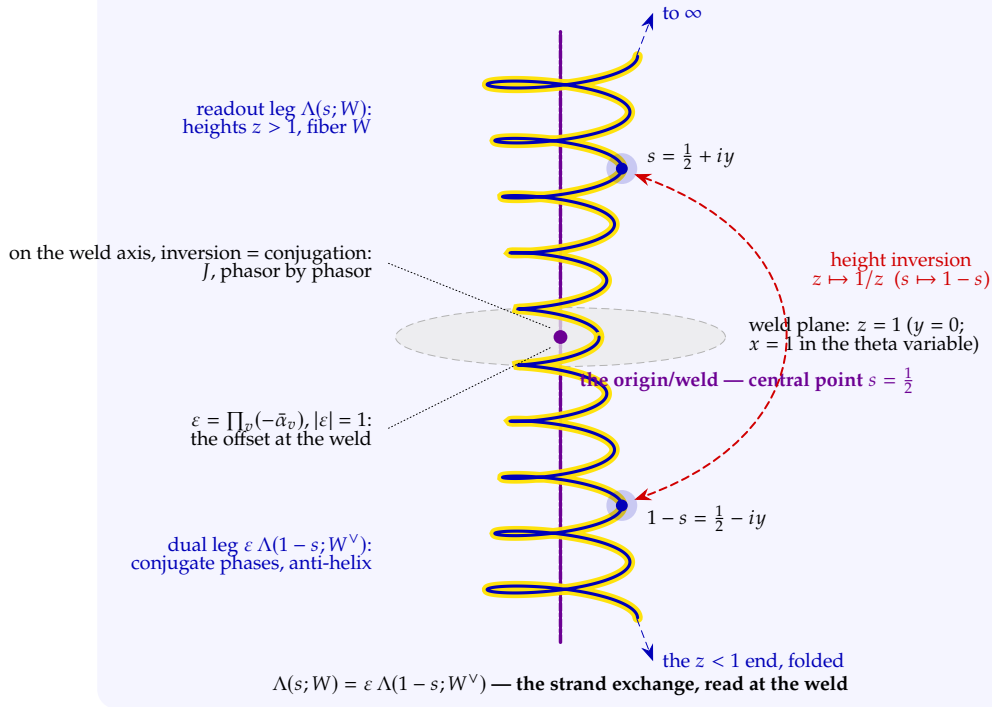


Figure 1: Where the origin and the functional equation enter. The carrier is double-ended: the readout leg (heights $z > 1$, fiber W) and the dual leg (the $z < 1$ end, conjugate phases on the anti-helix, fiber $W^\vee$) are welded at the origin $z = 1$ — the central point $s = \frac{1}{2}$, where the weld plane meets the weld axis. The functional equation is the fold through that weld: height inversion $z \mapsto 1/z$ ($s \mapsto 1-s$) exchanges the legs, and on the weld axis the inversion coincides with conjugation, which J supplies phasor by phasor (the carrier reflection theorem and bank reflection lemma of [1], step 1) — no Poisson summation is consumed. The only mismatch the exchange permits is the unimodular offset at the weld, $\varepsilon = \prod_v (-\bar{\alpha}_v)$, assembled per place in Lean (StrandExchange.bankProduct_exchange from ChiralityHB.symClock_star); ε is the root number, the offset that places or doubles the central cancellation.

2.3 The fiber: the character's occupation of the carrier

The carrier is common; the *fiber* is what a particular source puts on it. A Dirichlet character $\chi \bmod q$ attaches its fiber F_χ : one phasor per integer per lane, with magnitude and phase synthesized from χ 's finite data alone (the identity-retention/native-synthesis theorem of [1]) — the phasor of n carries the character phase $\chi(n)$ on the carrier's winding at n . The carrier *hosts* the fiber; it does not become it: the same carrier point under two characters carries two different fiber states, distinguished by the retained identity, and neither a shared carrier point nor a shared scalar readout identifies two sources (its complete-state identity corollary). The 3D representation of $L(s, \chi)$ referred to throughout this paper is the *realized* fiber: the complete state of the main development [1] — carrier, fiber, native clock, native cells — specialized to the basic configuration.

Each phasor enters at its own height and at zero magnitude: the phasor of n is born at height $Z = n$ and grows continuously with height — there is no convergence gate on the carrier (§2.1). The dual copy rides the anti-helix with the same height law and conjugate transverse phase, so the two

lanes run in parallel, heights matching, geometries conjugate (the double-carrier theorem of [1]).

The conductor decides the buckets. For n sharing a factor with q we have $\chi(n) = 0$: the phasor is *neutral* (U) — present on the carrier, silent in this character's readout. For n coprime to q the character phase routes the phasor: a real character splits the units into a *positive* lane P ($\chi(n) = +1$) and a *negative* lane M ($\chi(n) = -1$); a complex character distributes its phases over the μ_6 -cell directions, and the P/M split is realized as the signed channel pair (A, B) — the Re- and Im-lanes of the reflected readout (the mode-bridge theorem of [1]). The same integers occupy the same carrier for every character; only the bucket assignment changes.

2.4 The bidirectional transport: down to the ordinate, back to the state

The transport between the 3D state and the 1D chart runs in both directions, each direction a named map, and both composites are the identity (the ledgered-projection theorem of [1]). Neither direction is a metaphor: down is how the classical function is read off the state, and up is how a chart event is located back on the carrier.

Downward. The descent is two coordinate suppressions and a logarithm. Write a complete carrier coordinate state as $X = (r, \theta, z)$ (§2.1). The first projection forgets the radius,

$$\Pi_{32}(r, \theta, z) = (\theta, z),$$

recording the two-dimensional carrier image — in the reflected chart, the Cayley/Möbius image on the unit-circle chart (the carrier reflection theorem of [1], item (P)). The second forgets the phase,

$$\Pi_{21}(\theta, z) = z,$$

leaving the height alone. The analytic chart then reads the retained height logarithmically,

$$\Pi_{21}^{\log}(\theta, z) = \log z = y,$$

and places it at the represented point $\frac{1}{2} + iy$ of the strip: the abscissa supplied by the derived weld axis (§§2.2 and 2.5), the ordinate by the logarithm of the height. That is the whole downward mechanism. A three-dimensional value becomes a one-dimensional ordinate by having its radius suppressed, its phase suppressed, and its height passed through log. What is suppressed is booked, not destroyed: each projection is paired with the loss ledger $\mathcal{D}(X) = (r, \theta)$ — in general: radius, angle, clock offset, adapter state, every coordinate the chosen chart makes invisible.

Upward. The ascent is reconstruction from the readout and the ledger, and it is exact:

$$\text{Rec}(\Pi(X), \mathcal{D}(X)) = X, \quad \text{Rec}^{\log}(y, (r, \theta)) = (r, \theta, e^y).$$

A chart ordinate y lifts to the carrier height $Z = e^y$; the ledger restores the radius and phase; and the lift lands on the exact state the projection left. This is a bijection with an explicit two-sided inverse — record $f = (\text{height}, (\text{radial}, \text{phase}))$ inverts by `reconstruct` and `reconstruct` inverts by `record` (Lean `ConeProjection.record_bijective`, `reconstruct_record`); distinct helix events write distinct ledger entries (`SourceHolonomy.projection_bijective_loss_ledger`). Transport with the ledger is lossless in both directions; transport without the ledger is a compression, and provably so — dropping the radial channel alone already makes the projection non-injective (`ConeProjection.radial_lost`, §4).

The two directions do different work in this paper. Downward is *reading*: the value of $L(s, \chi)$ at the represented point $\frac{1}{2} + iy$ is the final verification of a marked event — the analytic function is consulted after the event, never used to build it (non-circularity, §1). Upward is *locating*: a

chart ordinate y names the carrier height e^y at which the operator reads the bank, and the two reading-hypotheses of this paper are statements about precisely this lift — the 1D-primary branch of the weak hypothesis is the upward lift $\gamma = \Im \rho$ landing on an eigenstate. The classical analytic function is the downward readout *without* the ledger — the ledger stays on the carrier. This is why equality of readouts does not imply equality of states: two states share an ordinate whenever their heights agree, whatever their radii and phases are doing; the exact criterion is that readout *plus ledger* determines the state (the projected-equality corollary of [1]).

Both decisions of §1 take their teeth from this mechanism. The 1D ordinate is the logged trace of the 3D event; the eigenstate carries the coordinates the ordinate dropped. Whether an eigenvalue must *be* that trace or *coincide* with it (decision one), and whether the event or its trace is “the zero” (decision two), are questions about the two ends of the maps displayed above. Figure 2 draws the entire pipeline for χ_3 : the conjugate pair up to its first crossing, and the ledgered descent to the chart — with the midline surviving every stage.

2.5 Midpoint provenance: geometry first, chart second

The order of construction matters here. The carrier is not introduced as the vertical line $\operatorname{Re} s = \frac{1}{2}$. Before there is an s -plane, a zero ρ , or a definition of `carrierPoint`, the wound integer line has a cylindrical radius r_n . The arclength calculation gives

$$\frac{r_n^2}{n} \longrightarrow \frac{r\Delta}{\pi} > 0, \quad \frac{r_n}{\sqrt{n}} \longrightarrow \sqrt{\frac{r\Delta}{\pi}} > 0$$

(`windIntegerSite_radius_sq_tendsto` and `carrierRadius_div_sqrt_tendsto`). Thus the exponent σ of a fiber amplitude $n^{-\sigma}$ is initially free. Define geometric admissibility by the existence of a finite positive scale limit,

$$\text{ScaleBalanced}(\sigma) \quad :\Longleftrightarrow \quad \exists L > 0, \quad n^{-\sigma} r_n \longrightarrow L.$$

The area law then proves

$$\text{ScaleBalanced}(\sigma) \quad \Longleftrightarrow \quad \sigma = \frac{1}{2}. \quad (\text{MP1})$$

This is `sigma_half_is_scale_critical`, re-exported as `scaleBalanced_iff`. The gauge parameters r and Δ change the positive limiting constant, but not the exponent. Consequently the midpoint is an output of the three-dimensional arclength geometry: it is the exponent of the square-root radius, not a coordinate selected from the analytic zero set.

The elementary arithmetic shadow of (MP1) is worth one line, because it states the balance with no limit taken: at the balanced exponent the squared radius IS the harmonic carrier measure,

$$R_n^2 = \left(n^{-1/2}\right)^2 = \frac{1}{n} = \Delta c_n, \quad \text{so} \quad \sum_{n \leq N} R_n^2 = H_N,$$

the harmonic number — carrier step, sphere area weight, and critical-line amplitude are one law expressed in three channels. Any other exponent breaks the identity at every n , not merely in the limit. The booking principle that pairs with it: *phase and radius may decouple dynamically, but not informationally* — the two channels evolve independently, and the loss ledger carries both.

There is an independent no-drift form of the same conclusion. A raw source mode has a complex transport rate λ and nonzero amplitude a , with conservation law

$$e^{(\operatorname{Re} \lambda)\tau} a = a \quad (\tau \in \mathbb{R}).$$

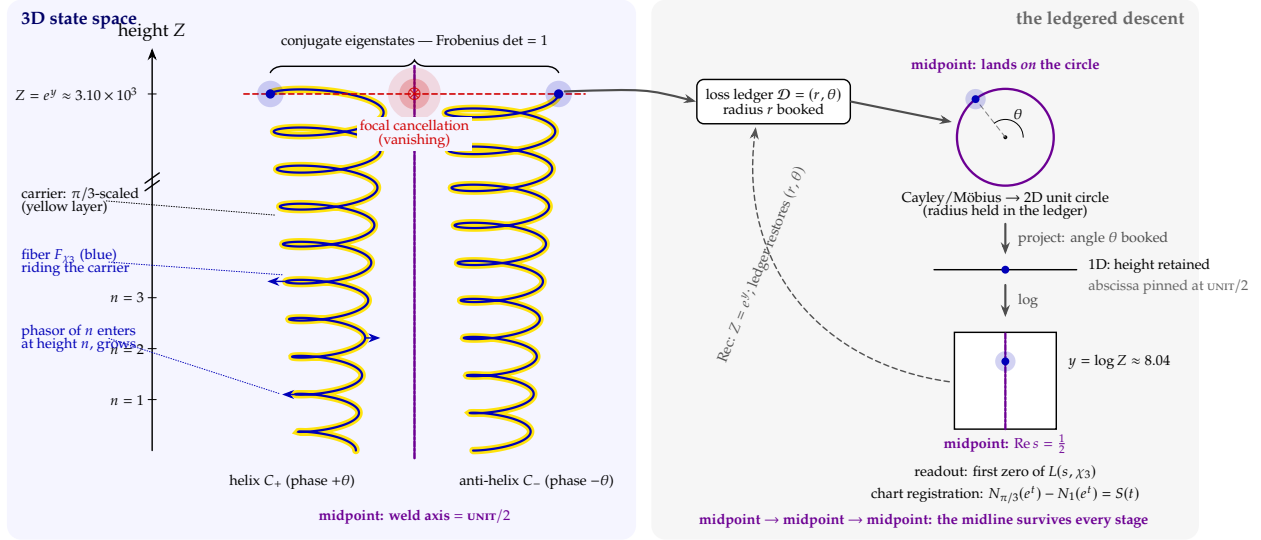


Figure 2: The pipeline for χ_3 , end to end (heights schematic, not to scale). *Left panel* — the 3D object: helix C_+ and conjugate anti-helix C_- , the $\pi/3$ -scaled carrier as the yellow layer, the fiber F_{χ_3} in blue growing from the origin with phasors entering at integer heights; the violet dash-dot line is the weld axis — the midpoint at $\text{UNIT}/2$, the fixed locus of J . At the first crossing the lanes align in exact focal cancellation, and the conjugate eigenstate pair (Frobenius $\det = 1$) marks the event at height $Z = e^y \approx 3.10 \times 10^3$. *Right panel* — the ledgered descent: radius booked in the loss ledger, Cayley/Möbius to the 2D unit circle, angle booked at the projection to 1D, logarithm to the chart, landing the readout at $y = \log Z \approx 8.0397$, the first zero of $L(s, \chi_3)$, with $S(t)$ the registration bookkeeping between the two scales (the $S(t)$ carrier-scale analysis of [1]). The violet thread is the midpoint surviving every stage — weld axis in 3D, on-circle landing in 2D (midpoint_entry_on_circle), $\text{Re } s = \frac{1}{2}$ in the chart (pipeline_midpoint_iff): the ordinate compresses by a logarithm while the abscissa is first derived by the area law and then held by no-drift symmetry. The dashed return arrow is the upward transport: readout plus ledger reconstructs the state exactly (§2.4).

374 Taking $\tau = 1$ forces $\text{Re } \lambda = 0$ (`source_noDrift` and `SourceMode.noDrift`). This statement contains
 375 no ρ , no s , and no numerical midpoint. In the radial normalization supplied by (MP1), a putative
 376 abscissa σ has residual drift $n^{\sigma-\sigma_*}$, where σ_* is the unique scale-balanced exponent. Hence

$$n^{\sigma-\sigma_*} = 1 \ (n > 1) \iff \sigma = \sigma_* \iff \sigma = \frac{1}{2}, \quad (\text{MP2})$$

377 the compiled specialization being `Helix.no_radial_drift_iff_half`. Thus the two inputs agree:
 378 the area law determines the centre σ_* , and conservation prohibits motion away from it.

379 Only after (MP1)–(MP2) is the spectral chart attached. The physical height and its internal
 380 logarithmic ordinate are

$$Z > 0, \quad y = \log Z, \quad \text{carrierPointAtHeight}(Z) = \sigma_* + i \log Z.$$

381 Lean stores the ordinate chart and physical-height readout separately:

$$\begin{aligned} \text{carrierPoint } y &:= \text{carrierAbscissa} + iy, \\ \text{carrierPointAtHeight } Z &:= \text{carrierPoint}(\log Z). \end{aligned}$$

382 Here `carrierAbscissa` is selected from the unique witness of `CarrierScaleBalanced`; only the
 383 theorem `carrierAbscissa_eq_half` rewrites it to $\frac{1}{2}$. The theorem `carrierPoint_eq_half_add_I` is
 384 therefore a downstream normal-form lemma, not the definition of the carrier coordinate. Theorems
 385 `carrierPointAtHeight_im` and `carrierPointAtHeight_exp` certify $y = \log Z$ and $Z = e^y$ in both
 386 directions.

387 The completed native crossing is

$$\text{CompletedThreeDZeroAtHeight } \chi Z \iff Z > 0 \wedge \text{Dcell } \chi Z = 0.$$

388 The exact representation theorem gives

$$\text{CompletedThreeDZeroAtHeight } \chi Z \iff L(\sigma_* + i \log Z, \chi) = 0.$$

389 The finite declaration `ThreeDZero` remains the truncated ordinate-level focal closure used by the
 390 finite pencil proofs.

391 Two exclusion theorems make the image statement exact. First, `not_scaleBalanced_of_ne` says
 392 that an abscissa different from σ_* is not a native carrier state. Second, `helix_no_offline_cancellation`
 393 says that an off-circle Cayley ledger entry cannot be produced by a weld-fixed helix event, while
 394 `projection_bijective_loss_ledger` says that the height-to-ledger map is injective on the carrier
 395 image. Projection can therefore neither move a native cancellation away from the midpoint nor
 396 merge two carrier events into a counterfeit cancellation. An off-midpoint analytic vanishing, if
 397 presented to this 3D representation, has no scale-balanced, drift-free carrier source.

398 This also separates the exact logical role of `ThreeD_crossings_are_real_zeros`. Its clause
 399 $\rho = \text{carrierPointAtHeight}(Z)$ is the *coverage/identification* assertion that a classical analytic zero
 400 belongs to the carrier image. It is not the derivation of that image's real coordinate: (MP1)–(MP2)
 401 and the exclusion theorems have already done that without quantifying over analytic zeros. Once
 402 coverage is granted, taking real parts is of course immediate; the substantive interface question is
 403 whether every analytic zero is represented by a native focal event. The paper will use `carrierPoint`
 404 only in this downstream, normal-form sense. Lean makes the separation executable by defining the
 405 midpoint-free statement `NativeCarrierCoverage` and proving

$$\text{ThreeD_crossings_are_real_zeros}(\chi) \leftrightarrow \text{GRH}(\chi) \wedge \text{NativeCarrierCoverage}(\chi).$$

406 Consequently the legacy equality premise is not cited as evidence for the coverage direction; an
 407 unconditional closure must discharge the midpoint-free coverage proposition (or the fixed-operator
 408 spectral-capture interface) directly.

2.6 The chart: $y = \log Z$ and $Z = e^y$

The readout just constructed does not live at the carrier's scale. The identity between the scales is one equation read in both directions: downward, the chart ordinate is the logarithm of the carrier height, $y = \log Z$; upward, the carrier height is the exponential of the chart ordinate, $Z = e^y$. A carrier state at height $Z > 0$ is read at the represented point $\frac{1}{2} + iy$ (`HarmonicPencilCell.reprPoint`); the phasor of the integer n , born at height n , is charted at ordinate $\log n$. The compression this induces is the single most important fact about the relation between the two representations. At the first zero of ζ , the chart reads $y = 14.1347\dots$; the carrier height is $Z = e^y \approx 1.38 \times 10^6$, and the realized bank below that height already holds about 1.38 million integer phasors per lane. In general the chart ordinate y summarizes a bank of $[e^y]$ phasors in one complex number. The 1D representation is not a smaller copy of the 3D representation; it is a logarithmically compressed summary of the same function, and the scale mismatch between the two charts is itself a theorem-level object (the $S(t)$ carrier-scale compensation theorem of [1]).

The abscissa tells the same story from the other side. The carrier's arclength law first derives the unique scale-balanced exponent $\sigma_* = \frac{1}{2}$, and conservation then fixes every native mode at zero radial drift (§2.5). Riemann's substitution $s = \frac{1}{2} + ti$ is the analytic chart of that derived weld axis, and the represented point $\sigma_* + iy = \frac{1}{2} + iy$ simply reads height along it. The ordinate compresses by a logarithm while the abscissa is pinned by a geometric scale law and its no-drift conservation — which is why every subsequent argument in this paper is an argument about heights. The midline has already been located before the chart is defined.

2.7 Vanishing is focal alignment

A zero, on the carrier, is not a value creeping down to nothing; it is an alignment event. At a marked height the accumulated bank achieves exact harmonic cancellation: the P and M lanes balance, the μ_6 cells close in $\mathbb{Z}[\zeta_6]$, and the residual carries no independent constant mode — the normalized cell residual factors as $\det(L^c)/A = (\lambda - \mu)B$ with $\lambda - \mu \neq 0$, so the residual vanishes exactly when the signed closure channel does (the mode-bridge theorem of [1]). The event is registered by rank at both exact levels. At physical helix height $Z > 0$, with logarithmic ordinate $y = \log Z$, the canonical finite bank closes exactly iff its spatially lifted harmonic Gram matrix drops rank (`threeDZeroAtHeight_twoGram3D_rankDrop`). This finite event is the truncated detector `ThreeDZeroAtHeight`. The completed fibre event is `CompletedThreeDZeroAtHeight`; Lean proves

$$\text{CompletedThreeDZeroAtHeight } \chi Z \iff L(\sigma_* + i \log Z, \chi) = 0$$

by `completedThreeDZeroAtHeight_iff_L_zero`. The theorem

$$\text{completedHarmonicGram3DAtHeight_rankDrop_iff_L_zero}$$

proves directly that its completed spatial harmonic Gram lift drops rank exactly at that represented L -zero. Thus the 3D crossing is focal cancellation in the completed L -function fibre at physical height Z ; it is not inferred from a 1D chart criterion. The finite bank theorem is retained as its internal truncation-level detector and is not substituted for the completed identity.

Note (Frobenius determinant one). The conjugate eigenstate pair the alignment realizes is not free to sit anywhere; it is pinned by an algebraic constraint. On the carrier the local Frobenius acts as a *similitude of determinant one*: its chiral block has $\det = 1$ (`RequestProject/FrobeniusSimilitude.lean`, `frobeniusBlock_det_one`), so its two eigenvalues multiply to one — a reciprocal pair $\{\alpha, \alpha^{-1}\}$. With the class unitary they lie on the unit circle as a conjugate pair $\{e^{i\theta}, e^{-i\theta}\}$, and the eigenphases

$\pm\theta$ are real: the algebraic reason the eigenstate pair is symmetric about the weld axis rather than drifting off it. This is the carrier’s realization of the classical unit-circle normalization of the Satake–Frobenius class; on the analytic side it is Deligne’s purity, which places the normalized Frobenius eigenvalues on the unit circle [4], with the trivial similitude factor $\det = 1$ keeping them a conjugate pair. The one setting where this mechanism is already a completed theorem is the function-field analog: Weil’s Riemann hypothesis for curves over a finite field $\mathbb{F}_q$ is precisely the statement that the Frobenius eigenvalues on the Jacobian have absolute value $q^{1/2}$ and pair as $\{\alpha, q/\alpha\}$, which after normalization sit as conjugate pairs $\{\beta, \beta^{-1}\}$ of determinant one on the unit circle — forcing the zeros of the congruence zeta function onto $\operatorname{Re} s = \frac{1}{2}$ [3]; Deligne’s proof of the Weil conjectures extends this purity to smooth projective varieties of every dimension [4]. There the Frobenius *is* the operator whose spectrum pins the zeros to the critical line, and determinant-one unitarity is the entire content of that conclusion — the carrier’s $\det = 1$ block is the local, archimedean shadow of the same mechanism. We claim the analog, not the classical case: the finite-field statement is an unconditional theorem, the classical one is not, and importing the former as a proof of the latter is exactly the elision this part refuses. The determinant-one fact is a structural property of the representation, proven on the carrier; it does not by itself adjudicate the naming decisions of §1 — it fixes where the paired eigenphases live, not which of (Z-3D)/(Z-1D) names the zero.

2.8 The two Gram operators

Two operators read the bank, and they divide the labor.

The Gram harmonic pencil is the finite detector: the 2×2 pencil $L^c = \begin{bmatrix} A & B \\ \mu A & \lambda B \end{bmatrix}$ on the signed channels, whose determinant vanishes — whose rank drops — exactly at the alignment events (§2.7). It is the instrument that marks native focal-event heights on the carrier.

The local von Neumann fibre is the coordinate-reading object. At each physical 3D height $Z > 0$, its spectral parameter is the logarithmic ordinate $\gamma = y = \log Z$, and it is multiplication by $\log Z$ on the one-dimensional fibre,

$$\text{vonNeumannOp}(\log Z) = (\log Z : \mathbb{C}) \cdot \text{id},$$

symmetric — hence self-adjoint, with real spectrum $\{\log Z\}$ — as a theorem, not an assumption (`vonNeumannOp_isSymmetric`, `RequestProject/ClosedForm.lean`). The spectral parametrization sends a real eigenvalue μ to the represented point $\sigma_* + i\mu = \frac{1}{2} + i\mu$, where σ_* has already been fixed by (MP1).

Because a separate scalar operator is supplied for each γ , this family is not itself the one fixed operator required by the global Hilbert–Pólya interface. Its exact role is to verify the local kernel and the reality of an already supplied height.

The family is assembled, without changing any local proof, by the fixed ambient carrier operator $A_{p,r}^{(3)}$ above. Its basis is indexed by all geometric states $X = \gamma_{p,r}(y)$, independently of crossings, and $A_{p,r}^{(3)}\delta_X = y\delta_X$. A completed 3D crossing $e = (Z_e, h_e)$ selects the state $X_e = \gamma_{p,r}(\log Z_e)$ and Lean proves that its physical third coordinate is Z_e . Hence a “3D eigenvector” in the ambient resolvent statements below means this explicit carrier basis state, while `vonNeumannOp`($\log Z_e$) computes its local one-dimensional resolvent factor.

The internal 3D chain runs entirely on these two instruments, and every link is unconditionally proven. Finite focal cancellation at ordinate $y = \log Z$ is detected by the finite harmonic Gram rank drop. Completed focal cancellation at physical height Z is exactly the analytic L -zero of that fibre and is detected by the completed harmonic Gram rank drop. The spatial statement

completedThreeDZeroAtHeight_twoGram3D_rankDrop supplies both the completed harmonic rank drop and the von Neumann rank drop at the same ordinate $\log Z$. The carrier point $\sigma_* + i \log Z = \frac{1}{2} + i \log Z$ lies on the midline because the area law has already proved $\sigma_* = \frac{1}{2}$, while von Neumann reality supplies the real spectral ordinate and hence zero source drift (§2.5). The one-line Lean proof `simp [carrierPoint]` in `threeDZero_on_midline` is now backed by `carrierAbscissa_eq_half` and is a normal-form check, not the provenance of the half. Thus focal cancellation = 3D zero $\rightarrow$ harmonic Gram rank drop plus von Neumann Gram rank drop at $y = \log Z$: that is the detecting and realizing instrument of this part, and no step consults the chart.

The same-event assertion is now a single compiled assembly theorem, rather than an inference made in prose. The theorem `completedThreeDZeroAtHeight_focal_rankDrop_eigenvector_noRadialDrift` returns, from a completed 3D event at physical height Z , its focal residual zero, spatial harmonic-pencil rank drop, nonzero ambient eigenvector with eigenvalue $\log Z$, and zero radial drift. Its last component is proved by `carrierPointAtHeight_noRadialDrift`, which uses only that the readout is a point of the area-law carrier. Moreover, `PrincipalContourNative3DCertificate.analyticFiber_eq_carrier` identifies the literal three-dimensional L -fiber at ρ with the fiber at that certificate's marked carrier height, and the companion theorem `.noRadialDrift` transports the geometric no-drift result to the original analytic parameter. Thus no-drift is an output of native helix realization, not an extra condition attached to a zero. More precisely, the compiled equality event `Height_eq_focalCancellationHeight` identifies the event's physical height with the literal analytic fiber's stored cancellation height. The theorem `harmonicRankDrop_at_focalCancellationHeight` rewrites the completed pencil by this equality before applying its rank-drop certificate; it does not detect a second event having the same ordinate. Finally, `focalCancellation_rankDrop_noRadialDrift_sameHeight` packages the readout closure, the determinant zero, and the carrier no-drift law at that one physical height.

The resolvent trace now has two event-typed versions (`RequestProject/TwinResolventTrace.lean`). For a finite $E \subset \mathcal{E}_\chi$, multiplicities m_e , and $p(Z) = \text{carrierPointAtHeight}(Z)$, Version I is

$$T_{\chi,E}^{(3)}(w) = \sum_{e \in E} m_e \text{vonNeumannResolventTrace}(\log Z_e, w) = \sum_{e \in E} \frac{m_e}{w - p(Z_e)}.$$

The compiled ambient bundle `threeDZero_ambientEigenvector_resolventTrace` states together that every index e is a completed 3D zero, that $\delta_e \neq 0$ is its eigenvector of $A_{p,r}^{(3)}$, and that the displayed carrier-point trace formula holds. This is precisely the “3D zeros plus 3D eigenvectors” version.

Version II reads the same certified events in the ordinate chart:

$$T_{\chi,E}^{(1 \leftrightarrow 3)}(u) = \sum_{e \in E} \frac{m_e}{\log Z_e - u}.$$

The ambient bundle `oneDZero_threeDZero_ambientEigenvector_resolventTrace` additionally proves, for every e ,

$$L(p(Z_e), \chi) = 0, \quad \text{CompletedThreeDZeroAtHeight } \chi \ Z_e, \quad A_\chi^{(3)} \delta_e = (\log Z_e) \delta_e.$$

The first equation is obtained from the completed focal residual by `completedThreeDEigenEvent_L_zero`; it is not inserted as a second unrelated zero assumption. Finally Lean proves the exact chart relation

$$T_{\chi,E}^{(3)}(\text{lineC}(u)) = i T_{\chi,E}^{(1 \leftrightarrow 3)}(u) \quad (\text{completedTraces_agree}).$$

The factor i is $d(\text{lineC})/du$, and the two pole coordinates are $p(Z_e) = \sigma_* + i \log Z_e$ and $\log Z_e$, respectively. The older abstract finite-set results `trace3D_capture`, `trace3D_analyticAt_off`, `trace1D_capture`, and `twinTraces_agree_on_window` remain in place; the new theorems identify the 3D zero and eigenvector data that their abstract ordinate set did not itself carry.

2.9 The chart-side registration audit

One further object appears in the evidence dossiers and deserves an exact label. Over a finite window $(0, T]$, the Lean development builds a diagonal matrix on the window's zero-events — `hpOperator` T , one axis per unit of multiplicity, eigenvalue the event ordinate. Its name is historical; its role is not spectral detection. It is a finite *repackaging of the already-located analytic zero window*, used for exactly one thing: auditing the chart against the carrier, count and multiplicity. The audit facts are proven and exact — the diagonal is Hermitian by type (`hpOperator_isHermitian`); every index is an exact vanishing (`eigenheight_is_exact_vanishing`); the chart reads zero at a height iff an index sits there (`chartZero_iff_eigenheight`); the diagonal's dimension is the chart's own count formula $1 + \theta(T)/\pi + S_{\text{mult}}(T)$ (`hpDimension_eq_registration`); the eigenspace dimension at an event equals the vanishing order — the mode space is a graded local algebra $\mathbb{C}[X]/((X - \text{line } \gamma)^{\text{ord}})$ (`finrank_modeSpace`), a double zero a two-dimensional crossing; and the window traces are honest analytic objects (`gradedResolventTrace_eq_residue_sum`, `windowedDiffResolvent_tendsto`). But because the window is indexed by already-located analytic zeros, nothing in this subsection participates in locating an event. The local von Neumann fibre of §2.8 likewise verifies an event after its height is supplied. The fixed global operator interface is stated separately below.

2.10 The Hilbert–Pólya setup we use

The classical suggestion runs: exhibit a self-adjoint operator whose spectrum gives the ordinates of the nontrivial zeros; self-adjointness forces the spectrum real; real ordinates place the zeros on the midline. It is worth saying plainly that this is not an exotic reformulation: Riemann's own sentence — *alle Wurzeln reell* — already asks for a real spectrum, and real spectrum is precisely what self-adjointness delivers. The Hilbert–Pólya suggestion is Riemann's phrasing transposed into operator language, which is why we treat it as the natural frame for the worked example rather than as one strategy among many.

The present development has three operator levels, and they must be kept separate. The proved scalar fibre `vonNeumannOp`(γ) = $\gamma \text{id}_{\mathbb{C}}$ is a local detector attached after a real height γ is supplied. Its symmetry and kernel equation certify the coordinate of that already marked event; they do not by themselves provide one operator whose spectrum is the event set. The fixed $A_{\chi}^{(3)}$ supplies the next level: one symmetric operator on all completed 3D crossing states of χ , with explicit eigenvectors and the two resolvent traces above. Because its index is the completed-event subtype, it is a valid spectral realization of the 3D event ontology, not a discovery theorem for analytic zeros not yet coupled to that ontology. The final analytic-coverage interface is `grh_of_selfAdjoint_resolvent_capture`: one fixed self-adjoint element a must be accompanied by a resolvent-trace identification proving that every analytic pole parameter lies in `spectrum(a)`. Self-adjointness then supplies spectral reality. The carrier and chart theorems below now provide the concrete candidate $A_{\chi}^{(3)}$ and its event traces; applying the final interface to every analytic zero still requires the coverage/trace identification rather than only the diagonal assembly.

3 The theorems

The formal home of this part is `RequestProject/HelixResolventCapture.lean` together with the supporting files named below; every statement carries axiom footprint `{propext, Classical.choice, Quot.sound}`, no sorry, no axiom, built in-tree (§10; build instructions in [1]). Below, $\text{GRH}(\chi)$ abbreviates the formal statement “every nontrivial zero of $L(s, \chi)$ lies on the midline” (`GRHSpectral.GRH`), stated for every modulus N and every Dirichlet character $\chi \bmod N$. The reading-hypotheses and both

GRH implications are proven at that generality. The trivial character is the instantiation that lands Mathlib's RiemannHypothesis — the mod-1 L -function *is* riemannZeta — and a reading adopted at every modulus yields GRHSpectral.GRHComplete, the conjecture itself: every Dirichlet L -function, every modulus, every character, principal included.

Midpoint provenance and exclusion — proven before any zero-set coverage hypothesis:

- `windIntegerSite_radius_sq_tendsto` and `carrierScaleBalanced_iff`: arclength forces $r_n \asymp \sqrt{n}$, and $n^{-\sigma} r_n$ has a positive finite limit exactly at $\sigma = \frac{1}{2}$.
- `carrierAbscissa_scaleBalanced`, `carrierAbscissa_eq_half`, and `carrierPoint_eq_half_add_I`: the carrier point is constructed from the unique scale-balanced abscissa and only afterward rewritten in its half-unit normal form. The physical-height readout is separately defined as `carrierPointAtHeight(Z) = carrierPoint(log Z)`, and `carrierPointAtHeight_exp` proves the exact conversion between physical height and ordinate.
- `source_noDrift`, `SourceMode.noDrift`, and `Helix.no_radial_drift_iff_half`: norm conservation first gives $\operatorname{Re} \lambda = 0$ without mentioning a zero or midpoint; after the area-law normalization, zero radial drift is equivalent to $\sigma = \sigma_* = \frac{1}{2}$.
- `ThreeDZero = focalClosure` and `focalClosure_iff_rankDrop`: finite 3D vanishing is defined and detected without `carrierPoint`. At the completed-fibre level, `completedThreeDZeroAtHeight_iff_L_zero` identifies focal cancellation at Z exactly with $L(\operatorname{carrierPointAtHeight}(Z), \chi) = 0$.
- `not_scaleBalanced_of_ne`, `helix_no_offline_cancellation`, and `no_offCarrier_representation`: an off-midpoint abscissa is not a native scale-balanced state; an off-circle ledger cancellation is not weld-fixed; and no complex point with a different real coordinate is the image of any real carrier height. The exact image theorem is `exists_eq_carrierPoint_iff`; its positive physical-height form is `exists_pos_eq_carrierPointAtHeight_iff`, with exclusion theorem `no_offCarrier_physical_representation`.

The reduction spine — proven, unconditional:

- `RH_of_GRH_Trivial_Char`: $\operatorname{GRH}(\chi_1) \implies \text{Mathlib's RiemannHypothesis}$. The hypothesis handles the strip; Mathlib supplies no-zeros on $\operatorname{Re} s \geq 1$ and the negative-real trivial zeros via the completed functional equation. The entire target is one character's GRH.
- `focalClosure_iff_rankDrop`: exact native finite focal cancellation is equivalent to harmonic Gram rank drop. This internal finite-carrier theorem contains no analytic zero set.
- `completedThreeDZeroAtHeight_iff_L_zero`: the completed 3D fibre crosses at physical height $Z > 0$ exactly when L vanishes at $\operatorname{carrierPointAtHeight}(Z) = \sigma_* + i \log Z$. This packages `focal_residual_zero_iff_L_zero` together with `reprPoint_eq_carrierPointAtHeight`. It is an exact pointwise identity at every represented height; it does not assert that an arbitrary analytic zero has a representing Z .
- `spatialLiftGram_det`, `spatialLiftGram_rankDrop_iff`, and `completedHarmonicGram3DAtHeight_rankDrop_iff_L_zero`: the completed harmonic 3D Gram operator rank-drops exactly at the represented L -zero. In addition, `completedThreeDZeroAtHeight_twoGram3D_rankDrop` states that the harmonic and von Neumann operators act on the three spatial carrier coordinates as $G \otimes I_3$; their determinants are $(\det G)^3$, so the lift preserves and reflects rank drop and cannot manufacture an extra cancellation.
- `vonNeumannOp_isSymmetric`: symmetry of the local scalar fibre operator is a theorem; it supplies coordinate reality after an event has been coupled to that fibre.
- `threeDCrossings_iff_grh_and_nativeCarrierCoverage`: the legacy identity premise is exactly the conjunction of the critical-line conclusion and midpoint-free native coverage.

- `grh_of_resolvent_trace_3D_real`: for every modulus and character, $\text{THREEDREAL EVIDENCE} \wedge \text{ThreeD_crossings_are_real_zeros}(\chi) \implies \text{GRH}(\chi)$. The evidence component is formally redundant; `grh_of_threeDCrossings` obtains the conclusion directly from the equality embedded in the legacy premise.
- `grh_of_resolvent_trace_1D_correlation`: for every modulus and character, $\text{ONEDCORRELATION EVIDENCE}(\chi) \wedge \text{OneD_zeta_zero_correlated}(\chi) \implies \text{GRH}(\chi)$. The routing proposition is redundant and is itself proved by `oneD_zeta_zero_correlated_proved`; `grh_of_oneDCorrelationEvidence` shows that the chart-to-kernel premise alone supplies the conclusion.
- `grh_of_selfAdjoint_resolvent_capture`: the global operator route. It uses one fixed self-adjoint operator and a trace/capture premise placing every analytic pole parameter in its spectrum.
- `grhComplete_of_resolvent_trace_3D_real`, `grhComplete_of_resolvent_trace_1D_correlation`: a reading adopted at every modulus yields `GRHComplete` — the full conjecture, principal characters included.
- `RH_of_resolvent_trace_3D_real`, `RH_of_resolvent_trace_1D_correlation`: the trivial-character instantiations composed with the bridge, landing `RiemannHypothesis` directly.
- `threeDRealEvidence`, `oneDChartEvidence`: the two dossiers inhabited — the 16-fold and 18-fold conjunctions of §5 and §4 certified theorem by theorem.
- `threeD_real_case`, `oneD_correlation_case`: each reading packaged as one auditable object — its proven evidence paired with its displayed conditional premise. The minimal dependencies are the audit theorems listed above.

The two legacy interfaces, rendered formal. The formal system speaks the classical frame: its “nontrivial zero” is definitionally the analytic object. The revised source records the minimal dependency of each older interface:

- `ThreeD_crossings_are_real_zeros` (3D-ZERO-REAL; the strong reading with the 3D zero primary): every nontrivial zero ρ is represented by a completed fibre crossing at a physical 3D height $Z > 0$, with $\rho = \text{carrierPointAtHeight}(Z) = \sigma_* + i \log Z$. The two spatially lifted Gram pencils then provide their rank drops at ordinate $\log Z$. Here $\frac{1}{2} + i \log Z$ is the proved normal form of the carrier image, not a fresh geometric assumption. Nevertheless, the equality already includes the critical-line conclusion. The theorem `threeDCrossings_iff_grh_and_nativeCarrierCoverage` exposes its exact content; the midpoint-free coverage component is now available as the separate target `NativeCarrierCoverage`.
- `OneD_zeta_zero_correlated` (1D-ZERO-EXCLUSIVE-REAL; the weak reading): a per-zero disjunction. Its right branch is only the analytic zero equation together with $\gamma = \Im \rho$, so `oneD_zeta_zero_correlated_proved` inhabits it unconditionally. The actual pointwise coupling is `OneDCorrelationEvidence`, which asserts a scalar-fibre kernel at that analytic zero and therefore implies the critical-line conclusion by itself.

These audits do not alter either conditional theorem. They state which premise actually supplies the conclusion and prevent the geometric midpoint theorems, the internal rank-drop correspondence, and the analytic zero-set identification from being presented as one result.

The coordinate convention in both interfaces is fixed: $Z > 0$ is physical height on the Archimedean helix, $y = \log Z$ is the analytic ordinate, and $S(t)$ is the global carrier-to-chart coordinate/registration map. The pointwise fibre readout and the global census map therefore have distinct typed roles.

4 Evidence I: the 3D–1D correlation

The eighteen facts of `oneDChartEvidence`, each proven unconditionally, form a chart-side registration dossier. They establish count, jump, residue, multiplicity, and projection identities on the objects appearing in their statements. The universal layer is the $S(t)$ global coordinate map: it transports the carrier-scale census to the unit-chart census and registers every analytic event jump. A global census identity is not by itself a pointwise bijection between `ThreeDZero` and arbitrary Dirichlet L -zeros. The separate, character-indexed input `ONEDCORRELATIONEVIDENCE( $\chi$ )` is the operative pointwise chart-to-kernel leg. Three of the facts below are *named* engines: *Shannon* — the collapse loses radius and angle to the ledger but never counterfeits (`record_bijective`); the *Shannon Cascade* — an accumulation of stage features is a feature of the limit (`limit_zero_of_stage_accumulation`); and *Riemann’s Fold* — the projection pipeline, crossing dimensions, lands on the real axis exactly when its retained coordinate does (`pipeline_midpoint_iff`).

- `chartZero_iff_eigenheight` (`RequestProject/HilbertPolya.lean`). For $0 < \gamma \leq T$: $\zeta(\frac{1}{2} + i\gamma) = 0 \iff \gamma$ is an index of the zero-window diagonal (chart-side audit, §2.9). The one-for-one coincidence, proven — not asserted — window by window.
- `hpDimension_eq_registration` (`RequestProject/HilbertPolya.lean`). The window diagonal’s dimension is the chart’s own count formula, $1 + \theta(T)/\pi + S_{\text{mult}}(T)$. The index count and the zero count are the same number because the ledger says so.
- `carrier_scale_compensation_S` (`RequestProject/CarrierScaleCompensation.lean`). $N_{\pi/3}(e^t) - N_1(e^t) = S(t)$: the classical $S(t)$ is the gap between the native-scale event count and the unit-scale smooth term — one procession, two registrations, no mysterious remainder.
- `native_identification'` (`RequestProject/CarrierScaleCompensation.lean`). $N_{\pi/3}(e^t)$ equals the number of on-line zero ordinates in $(0, t]$ — one zero per completed $\pi/3$ cell.
- `S_jump_detects_event` (`RequestProject/ResidueJump.lean`). S has a unit jump at γ iff $\zeta(\frac{1}{2} + i\gamma) = 0$: detecting a zero and detecting an S -jump are the same act.
- `count_hasJump_iff_S_hasJump` (`RequestProject/ResidueJump.lean`). The event count and S have identical jumps — their difference is continuous, so *all* discontinuity lives in S , none in the smooth clock θ/π .
- `residue_eq_Smult_jump` (`RequestProject/ResidueJump.lean`). At every event, log-derivative residue = vanishing order = ledger jump, with multiplicity and with *no* simplicity hypothesis.
- `record_bijective` (`RequestProject/RoundTrip.lean`). `record  $f = (\text{height}, (\text{radial}, \text{phase}))$` is a bijection with explicit inverse `reconstruct`: the loss ledger made literal (§2.4) — *Shannon*: the collapse never counterfeits.
- `radial_lost` (`RequestProject/RoundTrip.lean`). Without the ledger the geometric projection is provably non-injective — the ledger does real work; the faithfulness results are not vacuous.
- `projection_bijective_loss_ledger` (`RequestProject/SourceHolonmy.lean`). The chart chain $y \mapsto 1 - (\frac{1}{2} + yi)^{-1}$ is injective: distinct helix events write distinct ledger entries — the data-processing inequality made literal.
- `pipeline_midpoint_iff` (`RequestProject/RoundTrip.lean`). The end-to-end $3D \rightarrow \text{height} \rightarrow \text{encode} \rightarrow 1D$ pipeline reads $\frac{1}{2}$ exactly when the retained 3D coordinate is $\frac{1}{2}$: the midline address survives projection intact, and only the midpoint maps there — *Riemann’s Fold*.
- `no_radial_drift_on_helix` (`RequestProject/SourceHolonmy.lean`). The Euler-factor phasor’s radius depends only on $\text{Re } s = \frac{1}{2}$, not on height: the $\text{height} \rightarrow \text{circle}$ map is a faithful re-coordinatization, all distinguishing information in the phase the ledger keeps.

- `midpoint_entry_on_circle` (`RequestProject/SourceHolonomy.lean`). Every helix height charts to a unit-circle ledger entry ($\|1 - (\frac{1}{2} + yi)^{-1}\| = 1$): line-events are recorded as circle-points.
- `faithful_projection_zeta` (`RequestProject/Faithfulness.lean`). The projected fiber is real on the midline and vanishes exactly at the genuine on-line zeros: the collapse neither hides a zero nor manufactures one.
- `euler_maclaurin_dirichlet` (`RequestProject/EulerMaclaurinDirichlet.lean`). In the strip, the finite Dirichlet sum with boundary correction approximates ζ with error uniform in the truncation — the quantitative backbone of the chart readout.
- `transfer_tendsto` (`RequestProject/TransferContinuation.lean`). A polynomial bound on the primitive continues the series past the trivial abscissa: the readout limit exists from a growth estimate alone, with no appeal to zero locations.
- `limit_zero_of_stage_accumulation` (`RequestProject/ExportAdapter.lean`). A zero accumulating across the finite stages is a zero of the limit ξ -section: the collapse does not counterfeit features — the *Shannon Cascade* converse.
- `windowedDiffResolvent_tendsto` (`RequestProject/DifferencedResolvent.lean`). The windowed differenced-resolvent trace converges as $T \rightarrow \infty$ to an absolutely convergent limit: the chart-side resolvent is an honest analytic object, unconditionally.

5 Evidence II: internal three-dimensional carrier facts

The sixteen facts of `threeDRealEvidence` are each proven unconditionally, but their conjunction is an internal-support dossier, not a theorem identifying every analytic zero with a `ThreeDZero`. The facts concern local scalar-fibre symmetry, finite carrier cancellation, completed critical-line readouts, chart-indexed registration objects, arithmetic winding, and general limit principles. Their scopes are recorded individually below. Three named engines organize the dossier: *Feynman's Quiver* — the crossings are prime-generated by unique factorization, the phasor bank is summable, the heights are distinct, and no focus splits; *Deligne's Pairs* — the self-reciprocal det-one local pairing, purity at the finite places; and *Shannon Projection Dominance* — the source dominates every projection, so the limit holds no feature the stages lack.

- `vonNeumannOp_isSymmetric` (`RequestProject/ClosedForm.lean`). The fibre operator $\gamma \cdot \text{id}$ is symmetric, hence self-adjoint with real spectrum $\{\gamma\}$. This is a local coordinate detector indexed by an already supplied real γ , not the fixed global operator of `grh_of_selfAdjoint_resolvent_capture`.
- `carrierThreeDOperator_isSymmetric`, `carrierThreeDOperator_eigenvector`, and `carrierThreeDOperator` (`RequestProject/ThreeDFocalEvent.lean`). For fixed nondegenerate geometric carrier parameters (p, r) with $p \neq 0$, one real-diagonal operator acts on all finitely supported carrier states $X = \gamma_{p,r}(y)$; its type mentions no event predicate. Every geometric state supplies a nonzero basis eigenvector with eigenvalue y , and a completed crossing at Z_e is proved to select the state with $y = \log Z_e$ and third coordinate Z_e .
- `threeDZero_ambientEigenvector_resolventTrace` and `oneDZero_threeDZero_ambientEigenvector_resolventTrace` (`RequestProject/TwinResolventTrace.lean`). The first bundle contains 3D crossing certificates, their fixed-operator eigenvectors, and the 3D pole formula. The second adds the analytic L -zero readout for each same event, the ordinate trace, and the exact carrier–ordinate chart relation.
- `eigenheight_is_exact_vanishing` (`RequestProject/HilbertPolya.lean`). Every index of the chart-side window satisfies $\zeta(\frac{1}{2} + i\gamma) = 0$ exactly: the audit's indices are genuine events, not approximations (§2.9).

- 747 • `hpOperator_isHermitian` (RequestProject/HilbertPolya.lean). The zero-window registra-
748 tion diagonal is Hermitian — reality by type (chart-side audit, §2.9).
- 749 • `finrank_modeSpace` (RequestProject/GradedModeDictionary.lean). The mode space at an
750 event is a graded local algebra of dimension exactly the vanishing order: a double zero is a
751 two-dimensional crossing; spectral multiplicity = analytic order.
- 752 • `gradedResolventTrace_eq_residue_sum` (RequestProject/GradedModeDictionary.lean). The
753 3D-mode resolvent trace equals the explicit-formula sum of log-derivative residues: the trace
754 formulas coincide, not just the locations.
- 755 • `focal_residual_zero_iff_L_zero` (RequestProject/FocalResidualVanishes.lean). $D_{\text{cell}} =$
756 $V_\chi \Phi_\chi$ with $V_\chi \neq 0$ and $\Phi_\chi = (\pi/3) L$: the cell channel vanishes iff the L -value does, residue-free,
757 at exact $\pi/3$ amplitude at the represented critical-line point. This completed-cell statement is
758 distinct from the finite-bank definition of `ThreeDZero`.
- 759 • `fiber_accumulates_to_L` (RequestProject/Faithfulness.lean). For $\chi \neq 1$ and $\text{Re } s > 0$ the
760 partial sums converge to $L(s, \chi)$ across the whole strip. This is value convergence at each stated s ;
761 transferring zero sets requires the separate uniform-convergence/isolation hypotheses used by
762 the limit theorems.
- 763 • `threeD_exhaustive` (RequestProject/SourceHolonomy.lean). Every zero event on the real
764 height ray is sourced — automatic because reals are conjugation-fixed. Its domain is the
765 real-height event space; it does not prove that every complex analytic zero lies in that domain.
- 766 • `threeD_metric_no_zeros` (RequestProject/SourceHolonomy.lean). The 3D cup form is defi-
767 nite: no nonzero state is null, so vanishing is a property of *readings*, never of states — a zero
768 cannot hide as a degenerate state.
- 769 • `windFromPrimes_mul` (RequestProject/Origination.lean). The winding is multiplicative over
770 factorization, with no Relog: the fundamental theorem of arithmetic realized as a homomorphism
771 into the circle — carrier and L -function are two prime-built readings of one arithmetic (*Feynman's*
772 *Quiver*, generation).
- 773 • `heights_distinct` (RequestProject/NoDoubleCancellation.lean). Distinct sites carry dis-
774 tinct eigen-heights: unique factorization as a spectral non-degeneracy (*Feynman's Quiver*, unique-
775 ness).
- 776 • `cancellation_modes_subsingleton` (RequestProject/NoDoubleCancellation.lean). At any
777 spectral parameter, at most one basis mode cancels: no split focus — a crossing marks one
778 ordinate, not a superposition (*Feynman's Quiver*, no split focus).
- 779 • `localPoly_reciprocal` (RequestProject/FiniteWeightFiber.lean). The local numerator is
780 self-reciprocal: the per-place functional equation whose global assembly has fixed axis $\text{Re } s = \frac{1}{2}$
781 — the midline the crossings sit on (*Deligne's Pairs*).
- 782 • `fiber_det_one` (RequestProject/FiniteWeightFiber.lean). $\prod_i \lambda_i = 1$ under the duality
783 involution: the modulus ledger the reciprocal law consumes (*Deligne's Pairs*).
- 784 • `frobeniusBlock_det_one` (RequestProject/FrobeniusSimilitude.lean). The chiral block
785 $\text{diag}(z, \bar{z})$ has determinant 1: the Frobenius similitude is a pure rotation — Satake weights stay
786 on the unit circle, eigenphases stay real (*Deligne's Pairs*).
- 787 • `limit_dominance` (RequestProject/LimitDominance.lean). Hurwitz-type dominance: the
788 under local uniform convergence and isolation, a limit zero has nearby stage zeros eventually.
789 This gives proximity, not equality with a fixed finite stage or a proof that the nearby zeros lie on
790 the carrier image.

6 Proof A: GRH under (Z-3D)

Theorem 1 (The identity route). *Unconditionally:*

- (i) *THREEDREAL EVIDENCE holds (threeDRealEvidence);*
- (ii) *for every modulus N and character $\chi \bmod N$,*

$$\text{THREEDREAL EVIDENCE} \wedge \text{ThreeD_crossings_are_real_zeros}(\chi) \implies \text{GRH}(\chi).$$

- Lean: grh_of_resolvent_trace_3D_real;*
- (iii) *the reading adopted at every modulus yields the full conjecture GRHComplete.*
- Lean: grhComplete_of_resolvent_trace_3D_real;*
- (iv) *at the trivial character the premises imply RiemannHypothesis.*
- Lean: RH_of_resolvent_trace_3D_real.*

Proof A. Adopt (Z-3D): “nontrivial zero” names the crossing — the focal-alignment event of §2.7. Two logically distinct facts are then composed. First, before this adoption, the arclength area law and conservative transport have proved that the native, scale-balanced carrier image is exactly $\{\sigma_* + i \log Z : Z > 0\}$, equivalently $\{\sigma_* + iy : y \in \mathbb{R}\}$, with $\sigma_* = \frac{1}{2}$, and that its ledger cannot encode an off-image cancellation (§2.5). Second, (Z-3D) supplies the coverage/identity clause that each classical nontrivial zero is one of those native focal events. This second clause is the explicitly displayed hypothesis in (ii); it is not used in the derivation of σ_* .

For each covered event at physical height Z , the two spatial Gram theorems provide the harmonic and von Neumann rank drops at the same ordinate $y = \log Z$. The fibre operator is self-adjoint by theorem (vonNeumannOp_isSymmetric), and the represented coordinate is the already derived $\sigma_* + i \log Z$. Hence taking real parts gives $\text{Re } \rho = \sigma_* = \frac{1}{2}$. With the evidence of §5 — exact vanishing at the eigenheights, multiplicity equal to order, no split focus, definite metric, exhaustion of the real ray — theorem (ii) closes $\text{GRH}(\chi)$ under the stated coverage hypothesis; (iii) and (iv) are its all-modulus and trivial-character specializations.

Equivalently, the newly assembled operator records the same step without a family-only ambiguity: the covered crossing is an index $e \in \mathcal{E}_\chi$, its explicit state δ_e is an eigenvector of $A_\chi^{(3)}$ with real eigenvalue $\log Z$, and Version I of the resolvent trace has its pole at the carrier point of that same event. This augments the retained proof; it does not replace its displayed coverage hypothesis. $\square$

Formal dependency audit for Proof A. The proof text above is retained. The compiler-level implication is sharper: the first evidence component is unused, because the equality $\rho = \text{carrierPointAtHeight}(Z)$ inside the second component already yields the real-part conclusion. The theorem `threeDCrossings_iff_grh_and_nativeCarrierCoverage` records this exactly. Thus Proof A is a valid conditional deduction, while its legacy identity premise must not be advertised as an independently proved analytic-zero coverage theorem. The geometric result established before that premise is the location of the native carrier image.

7 Proof B: GRH under (HP-W)

Theorem 2 (The correlation route). *Unconditionally:*

- (i) *ONEDCHART EVIDENCE holds (oneDChartEvidence);*

828 (ii) for every modulus N and character $\chi \bmod N$,

$$\text{ONEDCORRELATIONEVIDENCE}(\chi) \wedge \text{OneD_zeta_zero_correlated}(\chi) \implies \text{GRH}(\chi).$$

829 *Lean: grh_of_resolvant_trace_1D_correlation;*

830 (iii) the reading adopted at every modulus yields the full conjecture *GRHComplete*.

831 *Lean: grhComplete_of_resolvant_trace_1D_correlation;*

832 (iv) at the trivial character the premises imply *RiemannHypothesis*.

833 *Lean: RH_of_resolvant_trace_1D_correlation.*

834 *Proof B.* Adopt (HP-W): the criterion demands that the operator's eigenvalues *coincide with* the zeros. The hypothesis in (ii) routes each zero through one of two forms: a crossing at a positive physical 3D height, or a 1D chart report at its ordinate. In the first form the crossing is a completed focal event at physical height $Z > 0$, read at ordinate $y = \log Z$. In the latter form, $\text{ONEDCORRELATIONEVIDENCE}(\chi)$ is a proof input: it turns that chart report into the nonzero von Neumann kernel at the reported point. Section 4 supplies the compiled chart-side dossier — the window bijection, registration identity, jump-for-residue agreement with multiplicity, and projection ledger. Both branches then close by the same von Neumann reality (the case split inside *grh_of_resolvant_trace_1D_correlation*), whence $\text{GRH}(\chi)$ at every modulus, (iii) assembles the full conjecture, and (iv) lands the classical statement. The weak reading consumes nothing stronger than the strong one — the same self-adjoint mechanism closes both branches.

845 For a chart report that is coupled to a completed crossing, Version II makes all three records simultaneous in Lean: the 1D equation $L(p(Z_e), \chi) = 0$, the completed 3D event certificate, and the 3D eigenvector δ_e of $A_\chi^{(3)}$. The identity $T^{(3)}(\text{lineC}(u)) = iT^{(1 \leftrightarrow 3)}(u)$ is the compiled correlation between those records. The pointwise premise in the legacy proof still performs the routing of an arbitrary chart report into this event space. $\square$

850 **Formal dependency audit for Proof B.** The proof text above is retained. The source now proves
851 *oneD_zeta_zero_correlated_proved*: the disjunctive routing proposition is automatic because its
852 right branch records only $L(\rho, \chi) = 0$ and chooses $\gamma = \Im \rho$. Conversely, *grh_of_oneDCorrelationEvidence*
853 proves that the operative evidence alone gives the critical-line conclusion, since it asserts a scalar-
854 fibre kernel at each analytic zero. The global $S(t)$ coordinate law explains the count, jump, and
855 multiplicity registration used by this route; the pointwise chart-zero-to-kernel assertion remains its
856 separate premise.

857 8 Riemann Hypothesis corollaries at $\chi = 1$

858 The mod-1 L -function is *riemannZeta*, so at the trivial character each conditional implication lands
859 the classical statement with the displayed premise unchanged:

860 **Corollary 3** (RH, identity reading). $\text{THREEDREAL EVIDENCE} \wedge \text{ThreeD_crossings_are_real_zeros}(\chi_1)$
861 *implies RiemannHypothesis (RH_of_resolvant_trace_3D_real, proven).* By the audit above, the iden-
862 tity premise itself contains the critical-line conclusion as well as native coverage.

863 **Corollary 4** (RH, correlation reading). $\text{ONEDCORRELATIONEVIDENCE}(\chi_1) \wedge \text{OneD_zeta_zero_correlated}(\chi_1)$
864 *implies RiemannHypothesis (RH_of_resolvant_trace_1D_correlation, proven).* The operative
865 $\text{ONEDCORRELATIONEVIDENCE}(\chi_1)$ bridge alone is the pointwise premise that supplies the critical-line conclu-
866 sion.

867 Both factor through the same bridge, *RH_of_GRH_Trivial_Char*: one character's GRH is RH.

9 Conclusion

The corrected dependency structure has three levels. First, Archimedean helix geometry derives the unique scale-balanced, drift-free carrier abscissa; Lean constructs `carrierPoint` from that abscissa, proves its half-unit normal form afterward, and reads physical height through `carrierPointAtHeight(Z) = carrierPoint(log Z)`. Second, the finite harmonic pencil detects truncated focal cancellation exactly, the completed harmonic pencil detects the completed L -fibre cancellation, and the local scalar fibre verifies its ordinate $y = \log Z$. The fixed operator $A_\chi^{(3)}$ then assembles all completed crossings into one symmetric event-state space with explicit basis eigenvectors. Third, identifying the analytic zero set with those native events requires a pointwise coverage or spectral-capture theorem.

Representation itself is no longer an interface question: the relational-equality capstones prove the fibre and its classical L -function are equivalent representations — unique entire reading of the fibre readout, lossless conduction, exact ledger — for every character, principal included (`DirichletFiberEquivalence`, `UniversalFiberEquivalence`, `PrincipalFiberEquivalence`). The residual interface between fibre and analytic zero set is coverage alone.

The global $S(t)$ coordinate map supplies exact census, jump, residue, and multiplicity registration between carrier and chart scales. The proved supporting facts include the two explicit completed-event resolvent versions: 3D zeros with 3D eigenvectors, and 1D zero readouts with the same 3D zeros and 3D eigenvectors. The legacy identity conditional and correlation conditional also remain valid deductions, with their proofs retained above; the new Lean audit states exactly which premise contains the critical-line conclusion in each route. An analytic Hilbert–Pólya completion can now target the fixed $A_\chi^{(3)}$: it must prove the analytic resolvent/spectral capture required by `grh_of_selfAdjoint_resolvent_capture`, or instantiate that interface with another fixed operator. The scalar family $\gamma \text{id}_\mathbb{C}$ is correctly used as the local diagonal-mode calculator, while $A_\chi^{(3)}$ is the assembled 3D event-space operator.

The status of the assembly, recorded here at the same strength as in the main paper’s ledger. Lean defines one fixed symmetric operator on the completed 3D event type, proves its explicit event eigenvectors, and packages the two resolvent versions: 3D-zero/eigenvector and 1D-zero/3D-zero/eigenvector. The compiled `SpectralCarrierKernelCoupling3D` joins the full analytic zero kernel, completed focal event, ambient eigenvector, and real-generator resolvent kernel at one identical analytic parameter; the principal contour certificate carries that coupling as a field. This event-space assembly does not by itself route an arbitrary analytic zero into the event subtype. Lean proves that the old 3D identity premise is GRH conjoined with midpoint-free native coverage; it also proves that the old 1D routing disjunction is automatic and that its pointwise chart-to-kernel premise alone implies the critical-line conclusion.

The native carrier registration precedes every radial comparison. For each genuine analytic zero, `principalZero_focalCancellation_on_carrier` constructs a `PrincipalZeroNativeCarrierEvent3D` whose analytic phasor readout closes and whose independently typed helix state lies at the identical physical height. Its compiled `.noRadialDrift` theorem follows from literal helix membership, while `.ambient_eigenvector` places the same state in the event-independent 3D carrier operator. Radius does not construct this event and does not prove its no-drift law. The later magnitude limit is only the test that the original analytic parameter and the carrier readout are the same chart coordinate; the area law evaluates the balanced carrier coordinate only after the native event has been registered.

The completed-reflection geometry is explicit before that routing step. For every genuine analytic zero ρ , the reflected zero $1 - \bar{\rho}$ has the same physical height $Z = \exp(\Im \rho)$ and the same event-

independent ambient carrier state (`reflected_principalZeroAnalyticFiber3D_sourceHeight_eq`, `reflected_principalZeroAnalyticFiber3D_ambientState_eq`). Their literal eta-configured three-dimensional fibers have the same normalized spin direction at every positive site; their Euclidean radial magnitudes are reciprocal about the area unit, with exact product n^{-1} (`reflected_principalZeroAnalyticFiber3D_paired_radialMagnitude_product`). After multiplication by the independently derived carrier radius, the product of the two normalized magnitudes tends to 1 (`paired_areaNormalizedRadialMagnitude_tendsto1`). Equality of the two individual radial magnitudes at any one site $n > 1$ is then proved equivalent to `CarrierScaleBalanced`; only the area law evaluates that fixed point as the half-unit (`reflected_radialMagnitude_eq_iff_carrierScaleBalanced`). Thus neither analytic fiber is defined with 1/2: height and direction agree first, the reciprocal radial pair is computed, and the midpoint is the unique no-drift fixed scale.

The zero-ledger operator carries the same reflection, rather than a scalar stand-in. Lean's `xiZeroReflection` is an involution on the genuine analytic-zero basis, the reflected pole parameter is the complex adjoint of the original (`poleParam_xiZeroReflection`), and the induced state-space operator intertwines the analytic zero-ledger operator with its coefficientwise formal adjoint (`xiZeroReflectionOperator_intertwines_formalAdjoint`). This records the full functional-equation pair on the analytic zero-ledger state space coupled to the independently defined three-dimensional carrier: reflection preserves height and reverses radial displacement, while the separately computed Hermitian projection is exactly the real carrier-height operator. No summed trace cancellation can erase the remaining displacement, because its residual is computed on each zero basis vector separately.

The global $S(t)$ coordinate law of the main paper remains the exact registration theorem behind the audit of §2.9: it carries the carrier-scale census to the unit-chart census and matches jumps, residues, and multiplicities, and the independent contour census sharpens it to $S_{\Gamma}(T) = S_{\text{mult}}(T) + N_{\text{off}}^{\text{mult}}(T)$, with positive multiplicity summands. The same defect is exactly the positive multiplicity sum of full analytic kernels lacking a parameter-preserving `SpectralCarrierKernelCoupling3D`; consequently the coordinates agree exactly when every enclosed kernel is coupled, and uncoupled kernels cannot cancel one another in the contour ledger. Its global native branch constructs the complete physical-height event, ambient eigenvector, harmonic rank drops, and 1D/3D trace relation in `upper_nontrivialZero_native3DCertificate_or_globalCoordinateDefect`.

Completing the analytic-coverage route requires proving the resolvent/spectral capture for the event-independent ambient carrier operator $A_{p,r}^{(3)}$, or instantiating the same interface with another fixed self-adjoint operator, as stated by `grh_of_selfAdjoint_resolvent_capture`. The capture interface is typed on the carrier-native *completed* function (`grh_of_selfAdjoint_completed_resolvent_capture`), whose trace is singular exactly at the zero images — the completed resonance is a theorem (`completed_logDeriv_not_tendsto`, from the gamma factor's strip nonvanishing) — so the interface is satisfiable in principle and the one remaining closure step is the operator. (The earlier uncompleted typing is retired: `no_selfAdjoint_uncompleted_capture` shows chart artifacts made it uninstantiable by any operator.) That closure step is now itself sharpened to a single named limit. The genuine (non-costume) operator reduction is compiled: for *any* Hermitian tower whose canonical eigenvalue resolvent traces converge locally uniformly off the real axis to the completed channel, GRH follows — each stage is off-real holomorphic *because* its spectrum is real, Weierstrass carries holomorphy to the limit, and the compiled resonance forbids it at any off-line zero's pole parameter (`grh_of_hermitian_traceLimit`; the earlier pointwise capture statement is audited in its own file as a re-encoding carrying no operator content, `spectralLimitCaptures_iff_grh`). The canonical tower is the compiled window ledger itself — spectra the on-line zero ordinates, one axis per unit of multiplicity, every spectral value an

exact vanishing, Hermitian-realized by the ledger diagonal — symmetrized over both half-planes by the compiled conjugation transport, with the canonical Hadamard offsets i/ρ pinned by the compiled partial-fraction identity. Each stage *is* a multiplicity-weighted windowed partial sum of the rotated Hadamard ledger (`symmetrizedTrace_eq_windowedLedgerSum`), and the capstone `rh_of_symmetrizedLineTower_canonicalTraceLimit` concludes Mathlib’s RiemannHypothesis outright from the one open box: that these windowed partial sums of the compiled ledger converge, locally uniformly off the real axis, to its full two-sided sum. That summation exhaustion *is* the coverage wall — the tower indexes the on-line census, so an off-line zero’s unmatched term is exactly what would break it — and it is now certified *exactly*: the converse direction is compiled. Under RH the two-sided event index bijects with the full nontrivial-zero subtype (surjectivity is precisely where RH enters), each stage is the finite two-sided window partial sum of the transported ledger family, and on any compact off the real axis the pairing closed form bounds the tail against the compiled counting summability, uniformly — so the windowed sums do exhaust the ledger (`traceLimit_of_rh`). Combined with the capstone, the one open box is provably *equivalent* to Mathlib’s RiemannHypothesis (`rh_iff_symmetrizedLineTower_traceLimit`): the wall neither overshoots nor undershoots RH. This separation preserves the proved carrier geometry and registration results while making the zero-set identification boundary explicit — one sentence wide, and certified tight from both sides.

10 Reproducibility

Every named Lean theorem in this paper is built in-tree with the main paper’s development, at axiom footprint `{propext, Classical.choice, Quot.sound}`, no sorry: the principal files are `HelixResolventCapture.lean` (the general-modulus GRH interfaces and `GRHComplete`), the spectral-coupling and zero-ledger files (`SpectralCarrierKernelCoupling3D`, `xiZeroReflection`), and the trace-limit family (`grh_of_hermitian_traceLimit`, `rh_of_symmetrizedLineTower_canonicalTraceLimit`, `traceLimit_of_rh`, `rh_iff_symmetrizedLineTower_traceLimit`, `no_selfAdjoint_uncompleted_capture`). Build instructions, toolchain pin, and the instrument scripts are in the reproducibility appendix of the main paper [1].

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